

To be submitted:

Doc file with answers to Qs listed below (write in your own words, all the textual data will be checked for AI-use, any flagging of AI-assistance will cause 0, even if the code works),

Code and its readme (it is better if you maintain a GitHub repo for your code, zip is also fine),

and one doc file that contains all your conversions with LLM (even if it 100 pages long, not a problem).

What is a good code:

Which runs in a single command and generates the required result (e.g. plot). If code execution is not automated, it won't be checked and hence, will be skipped for evaluation.

The readme should be very precise about execution command and the parameters. Even if you generate readme using LLM, double check everything before submission.

Assignment 0

Install PostgreSQL database engine (try to use the latest available version. Downgrade only for compatibility issue. Try not to use below 16) in your own machine, and load the JOB database (<https://event.cwi.nl/da/job/>).

ANALYZE all the tables so that the engine builds internal histograms.

Include the following details in your report under the title “**Setup**”:

Your machine details (CPU model and core specification, RAM and HDD amount, OS and PostgreSQL engine versions).

Include the following details in your report under the title “**Database**”:

Name: Join Order Benchmark (2013 Snapshot)

List down the table names, their PK column names and their cardinalities (run count(*) query for each of them to get to know the size)

Assignment 1: *Histogram*

Q1. Find out the histograms for columns “id” and “title” of the “title” table.

1. *What is the corresponding SQL?*
2. *What kind of histogram is it?*
3. *How many buckets are there?*
4. *Explain how textual attribute “title” is put into histogram.*
5. *List down the bucket boundaries.*
6. *Show the histogram as a bar-plot with value-boundaries in the X-axis and frequency in the Y-axis, as shown in the slides.*
7. *What is the max selectivity error for this histogram? Explain/show calculation.*
8. *How selectivity estimate is done for range query on text column such as “title”?*
9. *What is the estimated and actual selectivities of the queries:*

*select * from title where title.title < 'Race';*

*select * from title where title.id < '350000';*

Q2. Generate a code to create and visualize optimal serial histogram of 10-50 buckets (actual number is as per your choice) from column data fetched from the database. Now, use that code to create histograms for “id” and “title” columns of 3000 random samples drawn from the “title” table (use TABLESAMPLE command for sampling the table).

1. *Show the histograms as a bar-plot with value-boundaries in the Y-axis and frequency in the X-axis, as shown in the slides.*
2. *How much time did it take to build?*
3. *List down the frequency bucket boundaries.*
4. *What is the max selectivity error for this histogram? Explain/show calculation.*
5. *Repeat the sampling for 1000 and 5000, and build the corresponding histograms using the same code. Record the timings.*
6. *Using the time-data for 1000, 3000 and 5000 samples, extrapolate the time required to build histogram for the full table (you may use linear regression). List down the extrapolated times.*
7. *Create one plot to show the sample size vs time relationship.*
8. *Create one plot to show the sample size vs max selectivity error relationship.*
9. *If you choose less number of buckets, what will happen? What will happen for more number of buckets?*